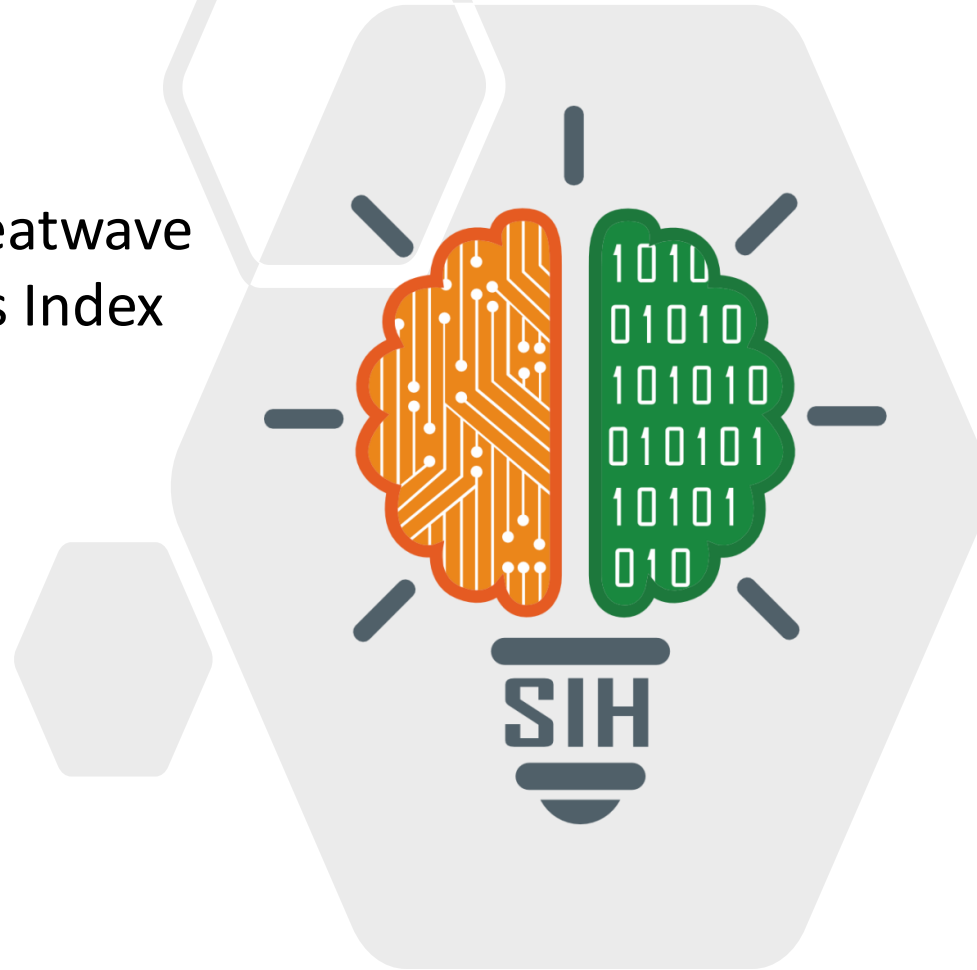


SMART INDIA HACKATHON 2026



- **Problem Statement ID** –SIH26083
- **Problem Statement Title**-Extreme Heatwave Early Warning and Human Thermal Stress Index
- **Theme**-Disaster Management
- **PS Category**- Software
- **Team ID**- 161620
- **Team Name**- DataShield





Problem Overview

- India's heat warnings rely mainly on **dry-bulb temperature**, overlooking *humidity, wind, and solar radiation* that drive actual heat stress.
- IMD heat-wave** thresholds are temperature-based, not **physiological** measures of *heat stress*.
- 40°C dry vs. humid heat** can cause very different health risks.
- No system links 3–5 day **hyperlocal weather forecasts** to predicted *hospitalizations or mortality rate*.



Proposed Solution

- ThermoSafe predicts heat stress **3-5 days** ahead using **UTCI, WBGT**.
- Ward-level **mortality** and **hospitalization alert** based on *demographics, housing, and health data*.
- It includes **color** coded **GIS dashboard** to guide cooling centers, health posts, and **ambulance placement**.
- ThermoSafe pushes timely **heatwave alerts** before the danger arrives, not after.

TEMPERATURE



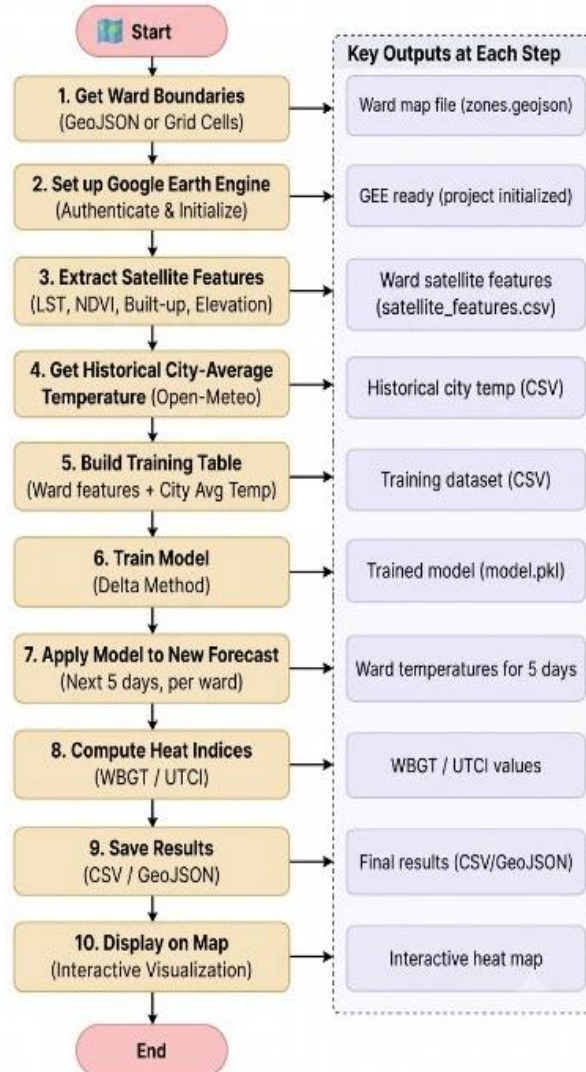
Uniqueness & Novelty

- Refined 'temperature feel'**: Additionally use vegetation or tree density (**NDVI**) and building density (**built-up%**) to measure the 'temperature feel'.
- Corrected air temp**: Uses **SUHI-to-air** scaling (unlike standard models).
- Comprehensive risk model**: Dual **mortality + hospitalization** tracking with transparent proxies.
- Ward-level insights**: Infers outdoor-worker density via OpenStreetMap where census data lacks.
- Self-calibrating thresholds**: Mortality/hospitalization cutoffs come from the city's own **historical summer data**, not another city's climate.
- Beyond Weather Apps**: ThermoSafe goes beyond coarse-grid estimates by using **ward-level satellite data on vegetation and built-up density to map heat risk** and drive targeted action.

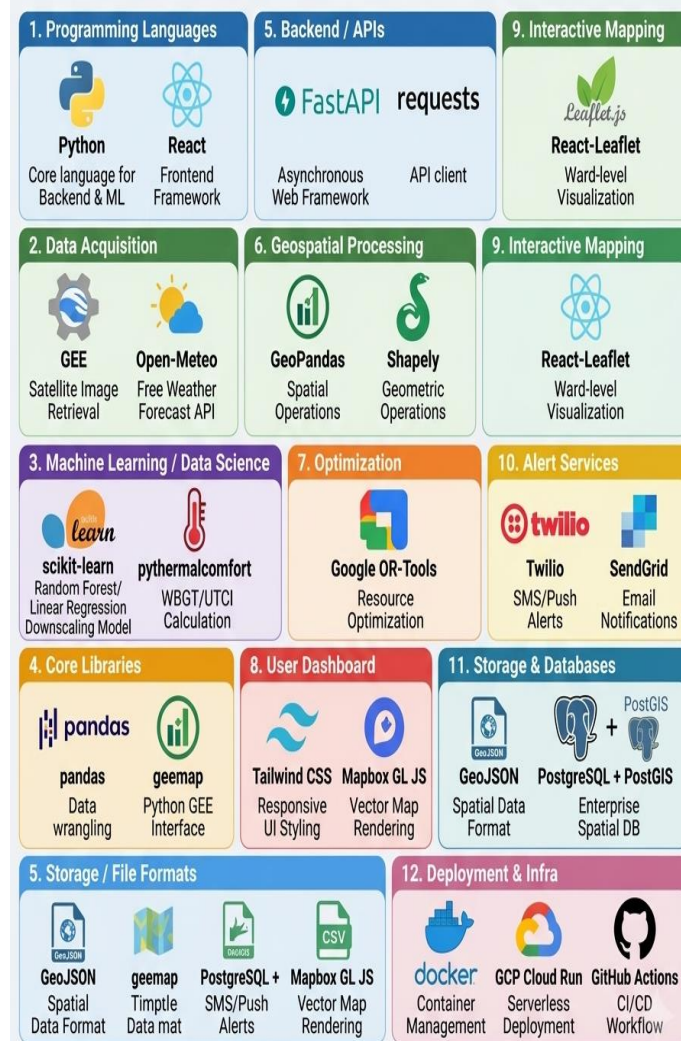
80-89°
FATIGUE POSSIBLE WITH PROLONGED EXPOSURE / PHYSICAL ACTIVITY
90°-104°
HEAT CRAMPS, HEAT EXHAUSTION POSSIBLE
105°-129°
HEAT EXHAUSTION LIKELY, HEAT STROKE POSSIBLE
130°+
HEAT STROKE HIGHLY LIKELY

TECHNICAL APPROACH

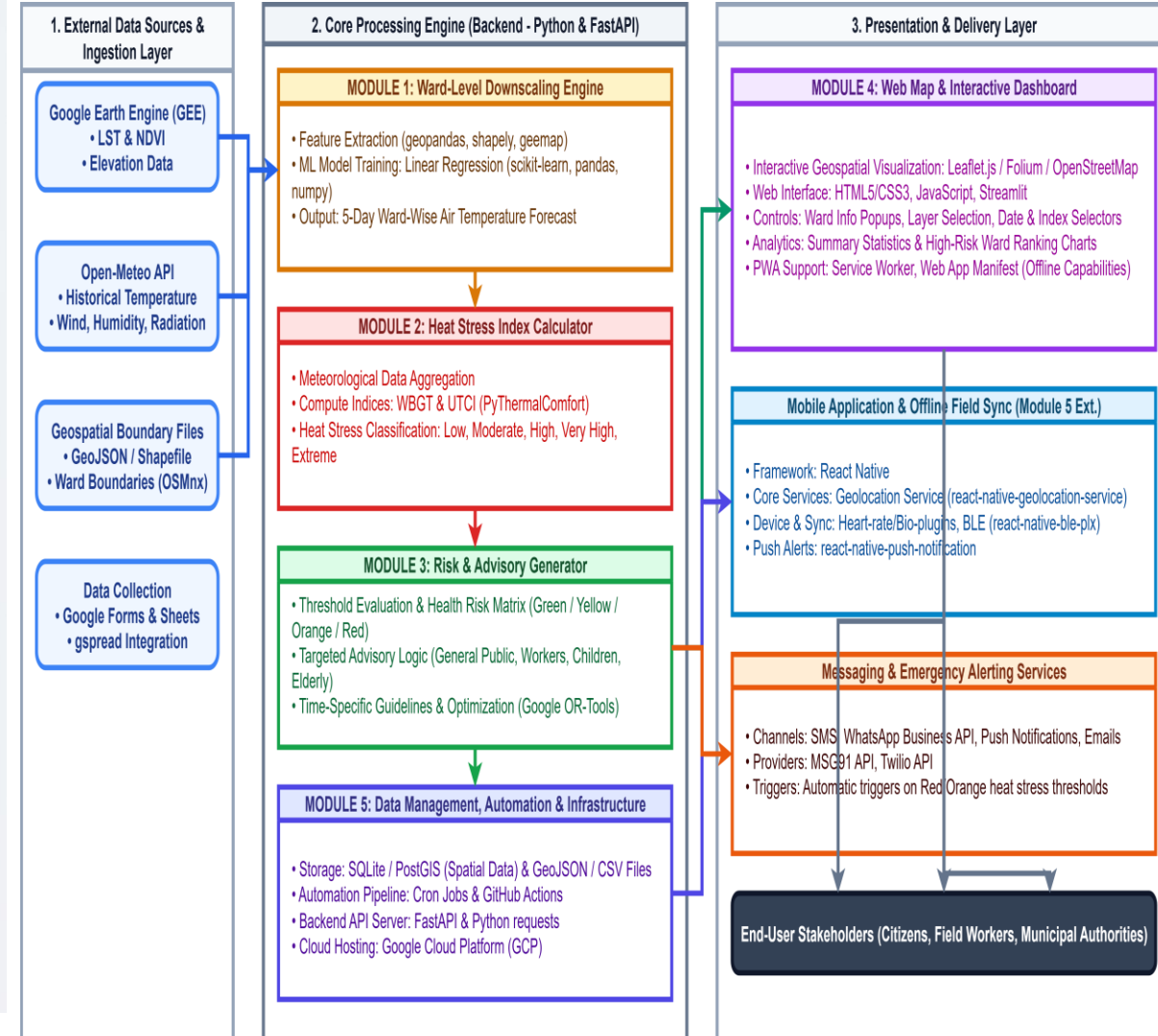
Low-level Flow Chart:



Tech Stack:



System architecture :





Feasibility of the Idea



Proven, Ready APIs

Open-Meteo, IMD, GEE & WhatsApp/SMS APIs plug in directly — no need to build from scratch.



Human-Centered Indices

UTCI & WBGT capture true physiological heat stress, not just air temperature.



Rich Public Data

Combines weather, satellite, population & vulnerability datasets.



Lean & Scalable

Open-source software & standard cloud infra GPU optional — the same pipeline expands new cities.



Mass-Reach Channels

SMS/WhatsApp alerts plus a citizen dashboard for wide accessibility.



Inclusive by Design

Local-language alerts & simple, actionable guidance for citizens, hospitals and authorities.



Potential Challenges & Risks



Data & Prediction Uncertainty

Weather/satellite gaps and AI model errors can affect ward-level accuracy.



Tech & Language Barriers

Patchy internet, low smartphone access, local dialects & code-mixing limit reach.



Privacy & Govt. Integration

Consent, secure data handling, API changes & coordination delays complicate rollout.



Strategies to Overcome Them



Continuous AI Validation

Explainable models, human-in-the-loop review & fallback alerts boost reliability.



Privacy-First Integration

Data minimization, encryption, role-based access & cached government data ensure a secure rollout.



Built for Low Connectivity

Offline/SMS support, multilingual NLP & gradual local calibration close the gap.

RISK

Risk: Satellite Cloud Cover & Data Gaps

Satellite availability is key, but but clouds block visual data.



Risk: Connectivity Issues in Vulnerable Wards

Unreliable internet/smartphone access in low-income areas limits reach.



Risk: API Rate Limits & Cost Overhead

Excessive API calls from GEE/Open-Meteo incur costs and service caps.



MITIGATION

Mitigation: Fallback to station-interpolated spatial kriging when satellite coverage is obstructed.



Use ground station network data as a reliable alternative.

Mitigation: Compressed SMS alerts and Progressive Web App (PWA) offline caching capabilities.

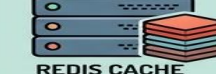


Deliver lightweight, text-based critical alerts.



Ensure app functions even without active internet.

Mitigation: Local Redis caching layer for forecast data to minimize external API calls.



Store data locally for rapid access and reduced bandwidth.

✓ Potential impact on the target audience



Citizens

Receive hyperlocal heat-risk alerts in advance with personalized safety guidance.



Local Communities

Access timely information in local languages for better awareness and preparedness.



Hospitals

Get early alerts to prepare for a possible increase in heat-related patients.



Healthcare Teams

Improve resource and staff planning based on predicted heat-related health risks.



Authorities

Gain ward-level risk visibility for data-driven heat-action planning and timely interventions.



✓ Benefits of the solution



Hyperlocal precision over blanket warnings

Replaces city-wide forecasts with ward-by-ward risk maps so dense wards get flagged even when the official forecast looks moderate.



Lives-based prioritization, not just heat maps

Folds in elderly population, housing vulnerability & health-insurance access to show which wards need help most.



Physiologically accurate risk

Uses UTCI/WBGT (humidity, wind, radiation) instead of raw °C — critical where 'feels like' differs sharply.



Quantified, defensible impact estimates

Outputs expected excess deaths per ward using peer-reviewed dose-response relationships for concrete planning.



Built on free, open & scalable infrastructure

Google Earth Engine, Open-Meteo, WorldPop & NFHS-5 — zero licensing cost, easily replicated across Indian cities.



Google Earth Engine



Open-Meteo



WorldPop



NFHS-5



Open Data
+
Open Tools



Urban Vulnerability, Micro-Climate & Downscaling

1. [Rajiva, A., et al. \(2021\). Assessment of urban heat island effects and thermal discomfort across major Indian cities. *Environmental Research*, 198, 111260.](#)
2. [Mahadevia, D., & Lathia, S. \(2020\). Testing urban vulnerability: Factors affecting informal settlements and slum density in Indian metros. *Down To Earth Report / CEPT University*.](#)

Extreme Heat, Mortality & Physiological Stress Metrics

1. [Azhar, G. S., et al. \(2014\). Heat-related mortality in India: Excess all-cause mortality associated with the 2010 Ahmedabad heat wave. *PLoS ONE*, 9\(3\), e91831.](#)
2. [Spangler, K. R., Liang, S., & Wellenius, G. A. \(2022\). Wet-Bulb Globe Temperature, Universal Thermal Climate Index, and other heat metrics for US counties, 2000–2020. *Scientific Data*, 9\(1\), 306.](#)

Market Research

Features	ThermoSafe	IMD Heat Bulletins	NWS HeatRisk	Commercial Apps (Tomorrow.io)
Hyperlocal Ward-Level Precision	✓	✗	✗	✓
Physiological Indices	✓	✗	✓	✗
GIS Emergency Resource Optimization	✓	✗	✗	✗
Self-Calibrating Local City Thresholds	✓	✗	✓	✗
Demographic & Worker Vulnerability Proxies	✓	✗	✗	✗
Low-Bandwidth & Offline Communication	✓	✗	✗	✗